

Abstract – Smart Career AI

Smart Career AI is a comprehensive, end-to-end career intelligence system designed to bridge the gap between an individual's current skill set and evolving job market demands. The platform leverages advanced Natural Language Processing (NLP) and machine learning techniques to automate resume analysis, job matching, and personalized career guidance.

The system begins with a resume parsing module that processes unstructured resume data and extracts critical information such as skills, educational background, and professional experience using NLP frameworks like spaCy and transformer-based models from Hugging Face Transformers. This structured representation of candidate profiles is then utilized by a job matching engine, which compares user profiles with job descriptions using similarity measures and ranking algorithms to generate a relevance score and identify best-fit opportunities.

Beyond direct job matching, the platform incorporates a career recommendation engine that suggests alternative job roles aligned with the user's existing competencies, enabling career exploration and transition planning. To support targeted skill development, a skill gap analyzer identifies missing or underrepresented skills required for both desired and recommended roles by benchmarking against industry datasets.

In addition, the system provides intelligent resume enhancement suggestions, including keyword optimization, content refinement, and Applicant Tracking System (ATS) compatibility improvements, thereby increasing the likelihood of candidate shortlisting. The integration of these modules creates a unified framework that not only evaluates candidate-job fit but also offers actionable insights for continuous career growth.

Overall, *Smart Career AI* functions as a personalized career assistant, empowering users to make data-driven decisions, improve their employability, and adapt effectively to dynamic job market requirements.